

Synthesis of New Chiral [2-(1-Hydroxyalkyl)pyrrolidino]methylferrocenes and Application to the Catalytic Asymmetric Addition of Diethylzinc to Arylaldehydes

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A series of new optically active [2-(1-hydroxyalkyl)pyrrolidino]methylferrocenes were conveniently prepared from commercially available *L*-proline and ferrocenecarboxylic acid. The crystal structure of key intermediate was obtained. They were then applied to catalyze enantioselective addition of diethylzinc to arylaldehydes with enantioselectivity up to 99%. The effects of the ligand structures on the enantioselectivity were also studied.

Keywords chiral [2-(1-hydroxyalkyl)pyrrolidino]methylferrocene, synthesis, asymmetric addition, diethylzinc, arylaldehyde

Introduction

Since the pioneering work of Hayashi *et al.*,¹ the design and synthesis of novel chiral ligands using ferrocene as a scaffold have received more attention to the asymmetric catalysis, with excellent asymmetric induction realized in many types of asymmetric reaction.² Recently, we reported the synthesis of chiral [2-(1-hydroxyalkyl)aziridino]methylferrocene ligands and their application to the enantioselective addition of diethylzinc to aldehydes.³ High yields and enantioselectivities have been achieved in this asymmetric addition reaction. The backbone of a chiral ligand should not be too flexible so as to provide an appropriate chiral environment. Therefore, ring compounds with certain rigidity are usually used as the framework.⁴ Although the backbone structures of chiral scaffolds are important to chiral induction, little is known about the real influence of chiral cyclic amino alcohols of different ring sizes on chiral induction.⁵ As part of a program aimed at the synthesis and application of chiral cyclic amino alcohols to asymmetric catalysis, in order to examine the effect of ring structure variations of the chiral ligands on chiral induction, in this paper, we present the synthesis of new chiral [2-(1-hydroxyalkyl)pyrrolidino]methylferrocenes from available *L*-proline and their use to the catalytic asymmetric addition of diethylzinc to arylaldehydes.

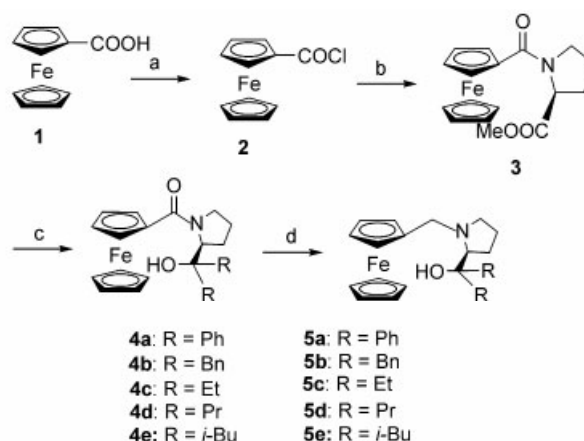
Results and discussion

Synthesis of chiral ligands 5

The synthesis of ligands **5a—5e** was shown in

Scheme 1. Ferrocenoyl chloride, generated *in situ* from ferrocenecarboxylic acid **1** and oxalyl chloride, was combined with *S*-proline methyl ester hydrochloride (prepared according to the literature method⁶) in the presence of Et₃N to form *S*-[2-methoxy carbonylpyrrolidino]carbonylferrocene **3** in 89% yield.⁷ Treatment of **3** with a range of Grignard reagents afforded *S*-[2-(1-hydroxyalkyl)pyrrolidino]carbonylferrocenes **4a—4e** in 54%—91% yields, and they were then reduced by LiAlH₄, affording ligands **5a—5e** in 85%—

Scheme 1



Reagents and conditions: (a) (COCl)₂, CH₂Cl₂; (b) *S*-proline methyl ester hydrochloride, Et₃N, CH₂Cl₂, two step yield 89%; (c) RMgX, THF, -10 °C to r.t., 54%—91%; (d) LiAlH₄, THF, 85%—93%.

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93% yield. The structure of **4a** was confirmed by X-ray single crystal analysis (Figure 1).

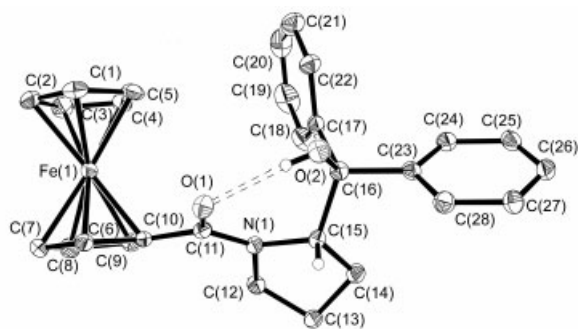


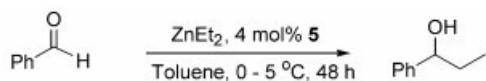
Figure 1 X-Ray crystal structure of **4a**. Selected bond lengths (nm) and angles (°) are: O(1)—C(11) 0.1243(3), O(2)—C(16) 0.1428(3), N(1)—C(11) 0.1349(3), N(1)—C(15) 0.1490(3), C(15)—C(16) 0.1560(3), O(1)—H 0.194(2), O(1)—O(2) 0.2640(3), O(2)—H 0.072(2); O(1)—C(11)—N(1) 121.2(2), N(1)—C(15)—C(16) 116.41(19), O(2)—C(16)—C(15) 108.91(18), O(2)—H—O(1) 165(3).

Application to asymmetric addition of diethylzinc to arylaldehydes

The catalytic asymmetric addition of diethylzinc to aldehydes is probably the most successful and still vigorously pursued area in asymmetric C—C bond formation as evidenced by hundreds of research papers and several comprehensive review articles.⁸ In addition, the reaction of diethylzinc with aldehydes has also become a classical test for the examination of the efficiencies of new ligands.

In order to clarify the chiral induction properties of these new [2-(1-hydroxyalkyl)pyrrolidino]methylferrocenes and differences of chiral induction between the three-membered ring and five-membered ring, the addition of diethylzinc to benzaldehyde, in the presence of 4 mol% chiral ligand **5** relative to benzaldehyde in toluene, was first investigated (Scheme 2). The results are listed in Table 1, and for comparative purposes, the results from corresponding [2-(1-hydroxyalkyl)-aziridino]methylferrocene ligands^{3c} are also listed in the corresponding entries.

Scheme 2 Addition of diethylzinc to benzaldehyde



As can be seen from Table 1, the addition of diethylzinc to benzaldehyde led to 1-phenylpropanol in 75%–87% yields with 26.3%–90.8% *ee*. It was shown that the enantioselectivity of the reaction was highly dependent upon the structure of chiral catalyst. The two ligands containing phenyl and benzyl groups afforded *S*-1-phenylpropanol in excellent yields of 87% and 83% with high enantiomeric excesses of 90.8% and 78.9% (Table 1, Entries 1 and 2). Ligand **5a**, with phenyl groups on the carbon bearing the hydroxyl group, showed a higher level of enantioselectivity than **5b** with

benzyl groups. But ligands **5c**–**5e** possessing dialkyl groups all gave 1-phenylpropanol in low enantiomeric excesses of 26.3%–30.3% (Table 1, Entries 3–5).

Table 1 Enantioselective addition of diethylzinc to benzaldehyde catalyzed by ligands **5a**–**5e**^a

Entry	Ligand	Yield ^b /%	<i>ee</i> ^c /%	Config. ^d	<i>ee</i> ^e /%	Config. ^e
1	5a	87	90.8	<i>S</i>	92.7	<i>S</i>
2	5b	83	78.9	<i>S</i>	—	<i>S</i>
3	5c	82	27.6	<i>R</i>	50.2	<i>S</i>
4	5d	75	30.3	<i>R</i>	18.7	<i>S</i>
5	5e	88	26.3	<i>R</i>	84.0	<i>S</i>

^a All reactions were carried out in toluene with the molar ratio of Et₂Zn/aldehyde of 2/1. The loading of **5** was 4 mol% relative to benzaldehyde. ^b Isolated yields. ^c Determined by HPLC on a chiral OD column. ^d Absolute configuration assigned by comparison with known elution order from a Chiralcel OD column according to the literature. ^e The *ee* and configuration were obtained via the corresponding [2-(1-hydroxyalkyl)aziridino]methylferrocene **6**.

It can also be seen that [2-(1-hydroxyalkyl)pyrrolidino]methylferrocenes **5a**, **5c** and **5e** gave lower *ee* values than the corresponding [2-(1-hydroxyalkyl)-aziridino]methylferrocene **6** (Figure 2, Table 1, Entries 1, 3 and 5). These results reflect the influence of change in different ring sizes of the ligand on the enantioselectivity. It could be believed that the backbone of three-membered ring was more rigid than that of five-membered ring and deduced that more rigid framework of [2-(1-hydroxyalkyl)aziridino]methylferrocene **6** led to the obvious increase of the enantioselectivity compared with [2-(1-hydroxyalkyl)pyrrolidino]methylferrocenes **5** in most cases.

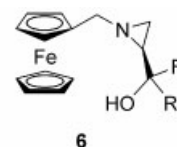


Figure 2 Structure of ligand **6**.

More interestingly, an unexpected reversal of configuration of the 1-phenylpropanol produced was observed when chiral ligands **5c**–**5e** were used even without changing the chirality of the ligand (Table 1, Entries 3–5). This may be due to the fact that the five-membered ring does not have the same degree of rigidity as three-membered ring. Moreover, the introduction of flexible dialkyl groups into the relatively flexible five-membered ring backbone may have made the chiral ligands more flexible, therefore, the 1-phenylpropanol product was formed via sterically different transition state. This change of product configuration is similar to that observed by Wang *et al.*⁹

Recently, Toma and coworkers reported that *S*-2-hydroxymethylpyrrolidinomethylferrocene **7** (Figure 3) catalyzed the asymmetric addition of diethylzinc to

benzaldehyde with 99% conversion but only 19.5% *ee*.¹⁰ Here it was noticed that the enantioselectivity was increased remarkably by putting substituents into the carbon atom bearing hydroxyl group of chiral ligand **7**, especially when two phenyls were introduced. Such results show that the enantioselectivity is very sensitive to the structure of ligand.

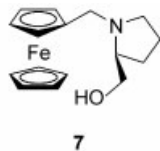


Figure 3 Structure of ligand **7**.

Consequently, the [2-(α -hydroxy- α -phenylbenzyl)-pyrrolidino]methylferrocene (**5a**) was chosen as the best chiral ligand for the further investigation into the effects of the ligand amount and temperature on the asymmetric induction, and the results are shown in Table 2. The best result was obtained with 5 mol% **5a** at 0 °C (Table 2, Entry 3). The reaction at room temperature (20–25 °C) led to a decrease in the enantioselectivity from 90.8% to 87.6%. The chiral ligand **5a** was recovered in excellent yield (*ca.* 96%) from the reaction mixture by preparative TLC and reused without any loss of enantiomeric purity of the product (Table 2, Entry 7).

Table 2 Asymmetric addition of diethylzinc to benzaldehyde with ligand **5a**^a

Entry	Temp./°C	Cat. mol/%	Yield ^b /%	<i>ee</i> ^c /%	Config. ^d
1	0	2%	84	66.9	S
2	0	3%	86	74.3	S
3	0	5%	95	90.8	S
4	0	7%	95	81.6	S
5	0	9%	93	81.6	S
6	r.t.	5%	88	87.6	S
7	0	5%	94	90.8 ^e	S

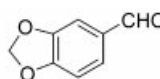
^a All reactions were carried out in toluene with the molar ratio of Et₂Zn/aldehyde of 2/1; ^{b–d} the same as those noted in Table 1; ^e the recovered ligand was used.

Chiral ligand **5a** was then examined for the asymmetric addition of diethylzinc to a series of aromatic aldehydes with all results showed in Table 3, and for comparative purposes, the results from the corresponding [2-(1-hydroxyalkyl)aziridino]methylferrocene were also listed.^{3c} A remarkable decrease of the *ee* values was observed with all tested arylaldehydes except *m*-chlorobenzaldehyde. These results reveal that the increase of the ring flexibility of cyclic amino alcohols led to the obvious decrease in the enantioselectivity of the reaction in most cases.

Conclusion

In conclusion, a series of new chiral [2-(1-hydroxyalkyl)pyrrolidino]methylferrocenes **5a–5e** were

Table 3 Asymmetric addition of diethylzinc to arylaldehydes catalyzed by ligand **5a**

Entry	Arylaldehyde	Yield ^a /%	<i>ee</i> ^b /%	<i>ee</i> ^c /%	Config. ^d
1	C ₆ H ₅ CHO	95	90.8	92.7	S
2	2-MeOC ₆ H ₄ CHO	88	57.4	93.5	S
3	4-MeOC ₆ H ₄ CHO	92	69.6	92.8	S
4		90	73.5	93.7	S
5	3-ClC ₆ H ₄ CHO	92	99.0	98.8	S
6	3-BrC ₆ H ₄ CHO	94	73.5	96.4	S
7	Formylferrocene	87	89.7	92.1	S

^a Isolated yield; ^b determined by chiral HPLC with chiracel OD column; ^c the *ee* value from the corresponding [2-(1-hydroxyalkyl)aziridino]methylferrocene ligand **6**; ^d absolute configuration assigned by comparison with known elution order from a Chiralcel OD column according to the literature.

conveniently synthesized from *L*-proline and ferrocene-carboxylic acid in four steps. The enantioselective ethylation of arylaldehydes with diethylzinc was investigated in the presence of a catalytic amount of the chiral ligands and the effects of the ligand structures on the enantioselectivity were then examined. The results showed that the increase of the flexibility of ligand structures led to the decrease of the reaction enantioselectivity and even reversal of configuration of the products. The further systematic studies on the influence of chiral cyclic amino alcohols of different ring sizes on chiral induction and the application of these ligands to asymmetric synthesis are in progress.

Experimental

General method

All reactions were carried out under nitrogen atmosphere condition and monitored by thin layer chromatography. Column chromatography purifications were conducted with silica gel. Melting points were determined on a YRT-3 melting point apparatus and uncorrected. Optical rotations were measured with a Perkin Elmer model 341 polarimeter at 20 °C. The *ee* values were determined by HPLC on a chiral column with hexane/propan-2-ol as eluent. The chromatographic system consisted of a JASCO model PU-1580 intelligent HPLC pump and a JASCO model UV-1575 intelligent UV/Vis detector (254 nm). The injection loop had a 20 μ L capacity. The column was operated at ambient temperature. NMR spectra of ¹H and ¹³C were performed on a Bruker DPX 400 MHz spectrometer in CDCl₃ or CD₃COCD₃ solutions referenced internally to Me₄Si with *J* values in Hz. IR spectra were determined on a Thermo Nicolet IR 200 spectrophotometer. Mass spectra were obtained on a Waters a-Tof microTM instrument with an electrospray ionization source (ESIMS). All the ESIMS measurements were performed utilizing MeOH as the solvent.

Reagents and solvents

Except for diethylzinc, which was purchased from Fluka, all other reagents were purchased from Beijing Chemical Reagent Company. Tetrahydrofuran and toluene were pre-dried over calcium chloride, and then distilled from sodium before use. Et₃N was dried with KOH pellets, then refluxed for 2 h with *p*-toluenesulfonyl chloride and distilled. Methylene dichloride was distilled from P₂O₅. All other reagents were commercially available and used as received. Benzaldehyde, 4-anisaldehyde, *m*-chlorobenzaldehyde and *m*-bromobenzaldehyde were distilled under a nitrogen atmosphere, respectively. 2-Anisaldehyde and heliotropine were recrystallized from hexane and ethanol, respectively.

Synthesis of S-[2-methoxycarbonylpyrrolidino]-carbonylferrocene (3)

To a stirred suspension of ferrocenecarboxylic acid (2.3 g, 10 mmol) in CH₂Cl₂ (30 mL) at room temperature under nitrogen was added via syringe oxalyl chloride (1.76 mL, 20 mmol). Gas evolution was accompanied by the formation of a dark red homogeneous solution after 10 min. The reaction mixture was stirred for an additional 30 min, followed by removal of the solvent *in vacuo*. The resultant crude ferrocenoyl chloride was dissolved in CH₂Cl₂ (20 mL) and added slowly to a solution of S-proline methyl ester hydrochloride (1.99 g, 12 mmol) and Et₃N (2.7 mL, 20 mmol) in CH₂Cl₂ (20 mL) at 0 °C under nitrogen. After stirring overnight the dark red reaction mixture was washed with H₂O (2 × 40 mL), dried over Na₂SO₄, filtered and evaporated *in vacuo*. The crude product was purified by column chromatography (1 : 4 to 1 : 2 of EtOAc/petroleum ether, V : V) to yield the crude compound 3. Recrystallization from EtOAc : hexane solution gave 3 as orange red needle crystals in 89% yield (3.03 g). m.p. 111.4–112.5 °C, $[\alpha]_D^{20}$ –73.0 (c 1.00, CHCl₃). ¹H NMR (400 MHz, CD₃COCD₃) δ: 1.91–1.97 (m, 1H, CH₂CHH), 2.06–2.16 (m, 2H, CHHCHH), 2.23–2.28 (m, 1H, CHHCH₂), 3.71 (s, 3H, OCH₃), 3.84–3.86 (m, 1H, NCHH), 3.88–3.90 (m, 1H, NCHH), 4.50–4.53 (m, 1H, NCH), 4.28–4.83 (m, 9H, FcH); ¹³C NMR (100 MHz, CD₃COCD₃) δ: 26.17, 48.94, 52.13, 61.07, 70.32, 70.67, 70.94, 73.13, 71.92, 77.13, 169.63, 173.75; IR (KBr) ν: 3111, 2923, 2853, 1740, 1593, 1467, 1406, 1360, 1198, 1165, 1100, 1031, 819, 765 cm^{–1}; HRMS (ESI) *m/z*: 341.0700 [M]⁺ (calcd for C₁₇H₁₉FeNO₃: 341.0714).

General procedure for the synthesis of S-2-(1-hydroxyalkyl)pyrrolidinocarbonylferrocenes (4a–4e)

To the Grignard reagent solution prepared from alkyl bromide (10 mmol) in THF (10 mL) and magnesium (0.24 g, 10 mmol) in THF (10 mL) was gradually added a solution of compound 3 (682 mg, 2.0 mmol) in THF (8 mL) at –10 °C over a period of 30 min. The mixture was stirred for 2 h in –10–0 °C, and then allowed to

reach the room temperature. After 24 h, the reaction was quenched with saturated aqueous NH₄Cl (15 mL) at 0 °C. The phases were separated and the aqueous phase was extracted with Et₂O (3 × 20 mL). The combined organic phases were washed with brine (20 mL), dried over Na₂SO₄, and after filtration the solvent was removed under reduced pressure. The residue was purified by the preparative TLC with CH₂Cl₂ : CH₃OH (97 : 3, V : V) or petroleum ether/EtOAc as developing solvent to afford 4a–4e.

S-2-(α-Hydroxy-α-phenylbenzyl)pyrrolidinocarbonylferrocene (4a): an orange red solid (848 mg, 91 %) by the preparative TLC with CH₂Cl₂/CH₃OH (97 : 3, V : V) as developing solvent. m.p. 164.1–165.9 °C (decompose), $[\alpha]_D^{20}$ –340.1 (c 0.4, CHCl₃); ¹H NMR (400 MHz, CDCl₃) δ: 1.35–1.41 (m, 1H, CH₂CHH), 1.53–1.62 (m, 1H, CH₂CHH), 1.88–2.01 (m, 2H, CH₂CH₂), 3.05–3.12 (m, 1H, NCHH), 3.53 (br, 1H, OH), 3.76–3.81 (m, 1H, NCHH), 4.10–4.80 (m, 9H, FcH), 5.27–5.35 (m, 1H, NCH), 7.28–7.60 (m, 10H, 2 × C₆H₅); ¹³C NMR (100 MHz, CDCl₃) δ: 24.46, 29.77, 51.09, 67.94, 68.08, 69.71, 69.94, 70.05, 72.88, 81.87, 127.19, 127.26, 127.46, 127.65, 127.93, 128.19, 143.74, 146.24, 173.65; IR (KBr) ν: 3436, 3135, 2966, 1582, 1466, 1403, 1249, 1166, 1107, 1029, 763, 703 cm^{–1}; HRMS (ESI) *m/z*: 465.1374 [M]⁺ (calcd for C₂₈H₂₇FeNO₂: 465.1391).

S-2-(1,3-Diphenyl-2-hydroxyprop-2-yl)pyrrolidinocarbonylferrocene (4b): an orange red solid (843 mg, 85%) by preparative TLC with petroleum ether/EtOAc (3 : 1, V : V) as the developing solvent. m.p. 124.3–126.1 °C (decompose), $[\alpha]_D^{20}$ +30.9 (c 0.4, CHCl₃). ¹H NMR (400 MHz, CDCl₃) δ: 1.54–1.61 (m, 1H, CH₂CHH), 1.85–1.93 (m, 2H, CH₂CH₂), 2.10–2.15 (m, 1H, CH₂CHH), 2.49 (d, *J* = 14.2 Hz, 1H, PhCHH), 2.78 (d, *J* = 13.5 Hz, 2H, PhCH₂), 2.98 (d, *J* = 13.2 Hz, 1H, PhCHH), 3.27–3.34 (m, 1H, NCHH), 4.04–4.09 (m, 1H, NCHH), 4.33–4.55 (m, 1H, NCH), 4.28–4.78 (m, 9H, FcH), 6.31 (s, 1H, OH), 7.17–7.43 (m, 10H, 2 × C₆H₅); ¹³C NMR (100 MHz, CDCl₃) δ: 25.33, 27.28, 42.04, 43.86, 51.39, 65.12, 68.39, 69.93, 70.10, 70.26, 70.35, 72.92, 76.72, 77.04, 77.11, 77.35, 79.47, 125.90, 126.14, 126.15, 127.89, 127.93, 128.40, 130.59, 130.59, 131.07, 137.65, 137.84, 174.17; IR (KBr) ν: 3457, 3221, 2917, 1575, 1459, 1406, 1159, 1106, 1031, 831, 755, 702 cm^{–1}; HRMS (ESI) *m/z*: 493.1730 [M]⁺ (calcd for C₃₀H₃₁FeNO₂: 493.1704).

S-2-(3-Hydroxypent-3-yl)pyrrolidinocarbonylferrocene (4c): an orange oil (472 mg, 67%) by preparative TLC with petroleum ether/EtOAc (3 : 1, V : V) as the developing solvent. $[\alpha]_D^{20}$ –16.3 (c 0.5, CHCl₃); ¹H NMR (400 MHz, CDCl₃) δ: 0.93–1.03 (m, 6H, 2 × CH₃), 1.26–1.74 (m, 6H, 2 × MeCH₂, CH₂CH₂), 1.85–1.93 (m, 1H, CH₂CHH), 2.08–2.12 (m, 1H, CH₂CHH), 3.39–3.42 (m, 1H, NCHH), 4.12–4.20 (m, 1H, NCHH), 4.55–4.60 (m, 1H, NCH), 4.25–4.84 (m, 9H, FcH), 5.72 (s, 1H, OH); ¹³C NMR (100 MHz, CDCl₃) δ: 7.51, 19.19, 22.99, 23.72, 25.47, 30.58, 51.55, 65.30, 65.58, 68.15, 70.49, 71.46, 73.16, 173.77; IR (KBr) ν: 3320, 3089, 2960, 2872, 1592, 1464, 1401,

1167, 1129, 1048, 1026, 819 cm^{-1} ; HRMS (ESI) m/z : 392.1295 $[\text{M} + \text{Na}]^+$ (calcd for $\text{C}_{20}\text{H}_{27}\text{FeNO}_2\text{Na}$: 392.1289).

S-2-(4-Hydroxyhept-4-yl)pyrrolidinocarbonylferrocene (**4d**): an orange solid (508 mg, 64%) by preparative TLC with petroleum ether/EtOAc (3 : 1, V : V). m.p. 76.1–77.9 °C, $[\alpha]_{\text{D}}^{20}$ –168.0 (c 0.5, CHCl_3); ^1H NMR (400 MHz, CDCl_3) δ : 0.92–1.05 (m, 6H, $2 \times \text{CH}_3$), 1.26–1.75 (10H, $2 \times \text{MeCH}_2\text{CH}_2$, CH_2CH_2), 1.88–1.91 (m, 1H, CH_2CHH), 2.06–2.09 (m, 1H, CH_2CHH), 3.38–3.41 (m, 1H, NCHH), 4.08–4.13 (m, 1H, NCHH), 4.51–4.55 (m, 1H, NCH), 4.23–4.85 (m, 9H, FcH), 5.89 (s, 1H, OH); ^{13}C NMR (100 MHz, CDCl_3) δ : 14.89, 15.27, 16.30, 16.34, 25.38, 27.37, 37.91, 40.25, 51.54, 65.75, 68.47, 69.89, 70.00, 70.30, 72.83, 77.41, 173.79; IR (KBr) ν : 3320, 3089, 2958, 2870, 1577, 1476, 1407, 1166, 1102, 1055, 820, 765 cm^{-1} ; HRMS (ESI) m/z : 397.1702 $[\text{M}]^+$ (calcd for $\text{C}_{22}\text{H}_{31}\text{FeNO}_2$: 397.1704).

S-2-(2,6-Dimethyl-4-hydroxyhept-4-yl)pyrrolidinocarbonylferrocene (**4e**): an orange solid (459 mg, 54%) by preparative TLC with petroleum ether/EtOAc (4 : 1, V : V) as the developing solvent. m.p. 139.6–141.2 °C, $[\alpha]_{\text{D}}^{20}$ –47.2 (c 0.5, CHCl_3); ^1H NMR (400 MHz, CDCl_3) δ : 0.89–1.06 (m, 12H, $4 \times \text{CH}_3$), 1.22–2.10 (m, 10H, $2 \times \text{Me}_2\text{CHCH}_2$, CH_2CH_2), 3.38–3.42 (m, 1H, NCHH), 4.00–4.10 (m, 2H, NCHH , NCH), 4.20–4.88 (m, 9H, FcH), 5.67 (s, 1H, OH); ^{13}C NMR (100 MHz, CDCl_3) δ : 19.19, 22.99, 24.20, 24.52, 25.41, 25.66, 25.81, 27.61, 29.46, 44.54, 46.37, 51.68, 65.93, 68.20, 69.73, 70.41, 73.05, 174.09; IR (KBr) ν : 3271, 3101, 1579, 1460, 1404, 1164, 1104, 1059, 823, 766 cm^{-1} ; HRMS (ESI) m/z : 426.2083 $[\text{M} + \text{H}]^+$ (calcd for $\text{C}_{24}\text{H}_{36}\text{FeNO}_2$: 426.2095).

General procedure for the synthesis of S-2-(1-hydroxyalkyl)pyrrolidinomethylferrocenes (**5a**–**5e**)

To a stirred suspension of LiAlH_4 (160 mg, 4 mmol) in dry THF (10 mL) at 0 °C, under nitrogen, was added a solution of compound **4** (1 mmol) in dry THF (10 mL) dropwise. After the reaction mixture was stirred for 5 h at room temperature and 4 h under reflux, wet Et_2O (10 mL) and saturated aqueous NH_4Cl (15 mL) were added carefully at 0 °C. The resultant gel was filtered through glass wool and washed carefully with Et_2O . The filtrate was dried over Na_2SO_4 , and the solvents were removed under reduced pressure. The resulting residue was further purified by recrystallization or preparative TLC with petroleum ether/EtOAc as the developing solvent to produce **5a**–**5e**.

S-2-(α -Hydroxy- α -phenylbenzyl)pyrrolidinomethylferrocene (**5a**): a yellow solid (419 mg, 93%), finally changeable by recrystallization from ethanol to needle crystals. m.p. 175.9–176.8 °C, $[\alpha]_{\text{D}}^{20}$ –114.0 (c 0.3, CHCl_3); ^1H NMR (400 MHz, CDCl_3) δ : 1.54–1.58 (m, 2H, CH_2CH_2), 1.69–1.71 (m, 1H, CHHCH_2), 1.87–1.90 (m, 1H, CHHCH_2), 2.31–2.32 (m, 1H, NCHH), 2.72 ($J=12.8$ Hz, 1H, FcCHH), 2.91 (d, $J=12.8$ Hz, 1H, FcCHH), 2.94–2.96 (m, 1H, NCHH), 3.81–3.84

(m, 1H, NCH), 3.82–4.02 (m, 9H, FcH), 4.83 (s, 1H, OH), 7.16–7.74 (m, 10 H, $2 \times \text{C}_6\text{H}_5$); ^{13}C NMR (100 MHz, CDCl_3) δ : 24.05, 29.74, 54.85, 55.00, 67.26, 68.23, 68.40, 69.50, 69.83, 69.91, 77.83, 84.31, 125.64, 125.89, 126.25, 126.28, 128.07, 128.15, 146.56, 148.47; (KBr) ν : 3202, 3085, 2973, 1488, 1448, 1394, 1101, 1060, 1026, 814, 743, 703 cm^{-1} ; HRMS (ESI) m/z : 452.1672 $[\text{M} + \text{H}]^+$ (calcd for $\text{C}_{28}\text{H}_{30}\text{FeNO}$: 452.1677).

S-2-(1,3-Diphenyl-2-hydroxyprop-2-yl)pyrrolidinomethylferrocene (**5b**): an orange red liquid (426 mg, 89%) by preparative TLC with petroleum ether/EtOAc (3 : 1, V : V) as the developing solvent. $[\alpha]_{\text{D}}^{20}$ +211 (c 0.5, CHCl_3); ^1H NMR (400 MHz, CDCl_3) δ : 1.63–1.79 (m, 4H, CH_2CH_2), 2.56–2.71 (m, 4H, $2 \times \text{PhCH}_2$), 2.84–2.87 (m, 1H, NCH), 2.99–3.06 (m, 2H, NCH_2), 3.40 (d, $J=13.3$ Hz, 1H, FcCHH), 3.64 (d, $J=13.3$ Hz, 1H, FcCHH), 3.83–4.05 (m, 9H, FcH), 7.19–7.34 (m, 10H, $2 \times \text{C}_6\text{H}_5$); ^{13}C NMR (100 MHz, CDCl_3) δ : 25.13, 27.53, 41.42, 42.91, 54.59, 57.27, 68.10, 68.17, 68.49, 68.55, 69.72, 69.94, 126.05, 126.12, 127.88, 127.97, 130.80, 131.05, 137.96; IR (KBr) ν : 3449, 3086, 2957, 1493, 1450, 1400, 1290, 1088, 1030, 819, 749, 703 cm^{-1} ; HRMS (ESI) m/z : 480.1989 $[\text{M} + \text{H}]^+$ (calcd for $\text{C}_{30}\text{H}_{34}\text{FeNO}$: 480.1990).

S-2-(3-Hydroxypent-3-yl)pyrrolidinomethylferrocene (**5c**): an orange red liquid (302 mg, 85%) by preparative TLC with petroleum ether/EtOAc (1 : 1, V : V) as the developing solvent. $[\alpha]_{\text{D}}^{20}$ –54.4 (c 0.6, CHCl_3); ^1H NMR (400 MHz, CDCl_3) δ : 0.83–0.95 (m, 6H, $2 \times \text{CH}_3$), 1.25–1.75 (m, 8H, $2 \times \text{MeCH}_2\text{CH}_2$), 2.62–2.65 (m, 1H, NCH), 2.86–2.89 (m, 2H, NCH_2), 3.51 (d, $J=13.3$ Hz, 1H, FcCHH), 3.76 (d, $J=13.3$ Hz, 1H, FcCHH), 4.11–4.24 (m, 9H, FcH); ^{13}C NMR (100 MHz, CDCl_3) δ : 7.79, 8.07, 25.19, 26.12, 27.19, 29.12, 54.68, 57.95, 67.89, 68.34, 68.57, 69.21, 69.68, 69.84, 75.73; IR (KBr) ν : 3442, 3093, 2965, 2876, 1460, 1401, 1129, 1104, 1030, 818 cm^{-1} ; HRMS (ESI) m/z : 356.1677 $[\text{M} + \text{H}]^+$ (calcd for $\text{C}_{20}\text{H}_{30}\text{FeNO}$: 356.1667).

S-2-(4-Hydroxyhept-4-yl)pyrrolidinomethylferrocene (**5d**): an orange red liquid (333 mg, 87%) by preparative TLC with petroleum ether/EtOAc (2 : 1, V : V) as the developing solvent. $[\alpha]_{\text{D}}^{20}$ –62.5 (c 0.5, CHCl_3); ^1H NMR (400 MHz, CDCl_3) δ : 0.89–1.00 (m, 6H, $2 \times \text{CH}_3$), 1.27–1.74 (m, 12H, $2 \times \text{MeCH}_2\text{CH}_2$, CH_2CH_2), 2.58–2.60 (m, 1H, NCH), 2.81–2.84 (m, 2H, NCH_2), 3.44 (d, $J=13.2$ Hz, 1H, FcCHH), 3.73 (d, $J=13.2$ Hz, 1H, FcCHH), 4.11–4.25 (m, 9H, FcH); ^{13}C NMR (100 MHz, CDCl_3) δ : 14.91, 14.95, 16.69, 17.10, 25.21, 27.20, 36.83, 40.06, 54.73, 58.03, 67.74, 68.27, 68.51, 69.57, 69.83, 75.51; IR (KBr) ν : 3447, 3096, 2957, 2871, 1460, 1398, 1133, 1104, 1000, 818 cm^{-1} ; HRMS (ESI) m/z : 384.1998 $[\text{M} + \text{H}]^+$ (calcd for $\text{C}_{22}\text{H}_{34}\text{FeNO}$: 384.1990).

S-2-(2,6-Dimethyl-4-hydroxyhept-4-yl)pyrrolidinomethylferrocene (**5e**): an orange red liquid (349 mg, 85%) by preparative TLC with petroleum ether/EtOAc (3 : 1, V : V) as the developing solvent. $[\alpha]_{\text{D}}^{20}$ –25.4 (c 0.5, CHCl_3); ^1H NMR (400 MHz, CDCl_3) δ : 0.92–1.02 [m, 12H, $2 \times \text{CH}(\text{CH}_3)_2$], 1.24–1.80 (m, 10H, $2 \times$

CH₂CH, CH₂CH₂), 2.52–2.58 (m, 1H, NCH), 2.82–2.89 (m, 2H, NCH₂), 3.42 (d, $J=13.2$ Hz, 1H, FcCHH), 3.76 (d, $J=13.2$ Hz, 1H, FcCHH), 4.11–4.33 (m, 9H, FcH); ¹³C NMR (100 MHz, CDCl₃) δ : 23.82, 24.15, 24.66, 24.74, 25.13, 25.21, 25.28, 27.63, 43.57, 46.35, 54.89, 58.40, 67.64, 68.12, 68.27, 68.50, 69.48, 69.68, 70.68, 85.70; IR (KBr) ν : 3427, 3096, 2953, 2868, 1465, 1398, 1105, 1041, 1001, 818 cm⁻¹; HRMS (ESI) m/z 412.2309 [M+H]⁺ (calcd for C₂₄H₃₈FeNO: 412.2303).

General procedure for the enantioselective addition of Et₂Zn to benzaldehyde

A solution of diethylzinc (1 mol/L in *n*-hexane, 1.1 mL) was added to a solution of the chiral ligand **5a** (0.025 mmol, 11.3 mg, 5 mol%) in dry toluene under nitrogen atmosphere. The mixture was cooled to 0 °C and stirred for 30 min. Freshly distilled benzaldehyde (0.05 mL, 0.5 mmol) was added to the mixture. The resulting mixture was stirred for 10 h at 0–5 °C and then allowed to warm to room temperature, and stirring kept for another 38 h at the same temperature. The reaction was quenched by the addition of saturated aqueous NH₄Cl (4 mL). The mixture was extracted with Et₂O (3 $\times$ 8 mL). The combined organic layers were washed with brine, dried over anhydrous Na₂SO₄, and evaporated under reduced pressure. Purification of the residue by the preparative silica gel TLC plate (hexane : EtOAc = 4 : 1) afforded the *S*-1-phenylpropanol. The *ee* was measured by HPLC on a chiral column.

X-ray crystallographic study

An orange red crystal of approximate dimensions 0.20 mm $\times$ 0.18 mm $\times$ 0.18 mm was mounted on a glass fiber. Crystallographic data of **4a** were measured on a Rigaku RAXIS-IV imaging plate area detector. The data were collected at 291(2) K with graphite monochromated Mo K α ($\lambda=0.071073$ nm), $2.05^\circ < \theta < 27.52^\circ$. The structures were solved by a direct method and expanded by Fourier techniques. The non-hydrogen atoms were refined anisotropically. Hydrogen atoms were included but not refined. All calculations were performed with the teXsan crystallographic software package. Crystal data of **4a**: orthorhombic $P2_12_12_1$: $a=1.1383(2)$ nm, $b=1.3570(3)$ nm, $c=1.4539(3)$ nm, $V=2.2459(8)$ nm³, formula C₂₈H₂₇FeNO₂ with $Z=4$, $D_{\text{calcd}}=1.376$ g/cm³, $F(000)=976$, $\mu(\text{Mo K}\alpha)=0.469$ mm⁻¹. Full-matrix least-squares refinement on F^2 based on 4778 independent reflections ($R_{\text{int}}=0.0273$) converged with 295 parameters. Final R indices [$I > 2\sigma(I)$]: $R_1=0.0349$, $wR_2=0.0704$ and R indices (all data): $R_1=0.0528$, $wR_2=0.0742$, with GOF=1.015.

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